DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III Topics in Analysis (3TN)

Tutorial Sheet 5 **August/September 2024**

1. Let $V = \mathcal{C}([0,1],\mathbb{R})$ be the space of continuous functions from $[0,1]$ to $\mathbb{R}$. Set

$$N_1(f) = \int_0^1 x|f(x)|dx \quad \text{and} \quad N_2(f) := \int_0^1 x^2|f(x)|dx \qquad \forall f \in V\,.$$

   (a) Show that N_1 and N_2 are norms on V.

   (b) Show that the norms N_1 and N_2 are not equivalent. (Hint: Consider the sequence of functions $f_n(x) = \dfrac{1}{x^2 + \frac{1}{n}}$)

2. Let

$$\mathcal{C}_0 := \left\{x = \{x_n\} : \lim_{n\to\infty} x_n = 0\right\} \quad \text{equipped with norm} \quad \|x\|_\infty := \sup_{n\in\mathbb{N}} |x_n|.$$

   (a) Prove that $(\mathcal{C}_0, \|\cdot\|_\infty)$ is a Banach space. (Hint: Knowing that $(\ell^\infty, \|\cdot\|_\infty)$ is a Banach space, we just need to show that $\mathcal{C}_0$ is a closed subset of ℓ^∞.)

   (b) Consider the mapping $L : \mathcal{C}_0 \to \mathbb{R}$ defined by

$$L(x) := \sum_{n=0}^{\infty} \frac{x_n}{n!} \qquad \forall x = (x_n) \in \mathcal{C}_0.$$

      (i) Prove that $L \in \mathcal{C}_0'$, the dual space of $\mathcal{C}_0$ with $\|L\|_{\mathcal{C}_0'} = \displaystyle\sum_{n=0}^{\infty} \frac{1}{n!}$.

      (ii) Prove that this norm is not achieved in $\mathcal{C}_0$. That is, there is no $x = (x_n) \in \mathcal{C}_0$ with $\|x\|_\infty = 1$ such that $L(x) = \displaystyle\sum_{n=0}^{\infty} \frac{1}{n!}$. **(Hint: by contradiction!)**

3. Let $\mathcal{C}^1([0,1])$ be the space of functions from $[0,1]$ to $\mathbb{R}$ which are continuous together with their first derivative. Set

$$H := \{f \in \mathcal{C}^1([0,1])\colon\ f(1/2) = 0\} \quad \text{and} \quad \sigma(f,g) := \int_0^1 e^x f'(x)g'(x)dx \quad \forall f,\, g \in H\,.$$

   Prove that σ is a scalar product on H.

4. Let $H = \mathbb{R}^3$ with the usual scalar product and $S := \text{span}((1,3,0),\, (0,1,1))$ (the subspace spanned by the vectors $(1,3,0)$ and $(0,1,1)$) and $u = (1,1,0)$, then:

   (a) show that $u \notin S$;

(b) find the orthogonal projection $P_S(u)$ of u onto S.

(c) Deduce, $d_E(u, S)$ where d_E is the Euclidean distance on $\mathbb{R}^3$.

5. Let $(V, \|\cdot\|)$ be a normed space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Show that the norm $\|\cdot\|$ satisfies the parallelogram identity if and only if

$$\|u - v\|^2 + \|u + v\|^2 \leq 2(\|u\|^2 + \|v\|^2) \qquad \forall u,\, v \in V\,.$$

6. Let $(H, (\cdot, \cdot))$ be a Hilbert space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let S, S_1, S_2 be subspaces of H. We recall that for every $A \subset H$, the orthogonal complement $A^\perp$ of A is defined as

$$A^\perp := \{u \in H\colon\ (u, v) = 0 \quad \forall v \in A\}\,.$$

(a) Let A, B be two subsets of H such that $A \subset B$. Show that $B^\perp \subset A^\perp$.

(b) Show that $S^\perp = \overline{S}^\perp$, where $\overline{S}$ denotes the closure of S in the normed space $(H, \|\cdot\|_H)$ with $\|\cdot\|_H$ being the norm induced by the scalar product $(\cdot, \cdot)$.

(c) Prove that $(S_1 + S_2)^\perp = S_1^\perp \cap S_2^\perp$.

(d) If S is a closed subspace, then show that $(S^\perp)^\perp = S$.

(e) Deduce from (b) and (d) that $(S^\perp)^\perp = \overline{S}$.

7. Let $\mathbb{R}^4$ be equipped with the usual dot product. Let

$$F := \{(x, y, z, t) \in \mathbb{R}^4\colon\ x + y + z = 0 \text{ and } x + y + z + t = 0\}\,.$$

Find a basis of F and a basis of $F^\perp$.